

Trigonometry Most Repetitive questions

Some Extra Results

$a^2+b^2=(a+b)^2-2ab$

$a^3+b^3=(a+b)^3-3ab(a+b)$

$$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta \cdot \cos^2\theta$$

$$\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta \cdot \cos^2\theta$$

$(\sin^2\theta)^2+(\cos^2\theta)^2 = (\sin^2+\cos^2)^2 - 2\sin^2\cdot\cos^2$

$\sin\theta \times \cos\theta$: $\frac{1}{\csc}$ $\frac{1}{\sec}$ $\csc\times\sec$

$$(\sin\theta \pm \cos\theta)^2 = 1 \pm \sin 2\theta$$

$1+2sc$ $1-2sc$

$$\tan\theta + \cot\theta = \frac{1}{\sin\theta\cos\theta}$$

$\frac{s}{c}+\frac{c}{s}=\frac{s^2+c^2}{sc}$

$$\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$$

$$\cos^4\theta - \sin^4\theta = \cos^2\theta - \sin^2\theta$$

$$\sec^4\theta - \tan^4\theta = \sec^2\theta + \tan^2\theta$$

$$\operatorname{cosec}^4\theta - \cot^4\theta = \operatorname{cosec}^2\theta + \cot^2\theta$$

$$\tan^2\theta - \sin^2\theta = \tan^2\theta \times \sin^2\theta$$

$$\cot^2\theta - \cos^2\theta = \cot^2\theta \times \cos^2\theta$$

$1+2sc = \frac{1}{9} - 1 = +\frac{8}{9}$

| If $sinA - cosA = 1/3$ and $A \in (0, \pi/2)$, then what is $\sin 2A$? → 2s·c

यदि $sinA - cosA = 1/3$ और $A \in (0, \pi/2)$ है, तो $\sin 2A$ क्या है?

a) $\frac{2}{3}$

b) $\frac{4}{9}$

c) $\frac{8}{9}$ ✓

d) $\frac{1}{2}$

PARMAR SSC

2 **The value of $\sin^4\theta + \cos^4\theta$ is equal to:**

$\sin^4\theta + \cos^4\theta$ का मान बराबर है:

a) 1

b) $1 - 2\sin^2\theta\cos^2\theta$

c) $\sin^2\theta + \cos^2\theta$

d) $2\sin^2\theta\cos^2\theta$

PARMAR SSC

① $1 - 2s^2c^2$

$= 1 - 2 \times \frac{81}{1024}$

$\frac{81}{512}$

3 If sinθ + cosθ = 5/4, and θ is acute, what is the value of $\sin^4\theta + \cos^4\theta$?

यदि sinθ + cosθ = 5/4, तथा θ न्यून कोण है, तो $\sin^4\theta + \cos^4\theta$ का मान क्या है?

a) 225/16

b) 431/512

c) 677/85

d) 239/18

$1 + 2sc = \frac{25}{16} - 1$

$sc = \frac{9}{32}$

PARMAR SSC

$1+2sc = \frac{3}{4} - 1 = -\frac{1}{4}$

$1-\left(-\frac{1}{4}\right) = 1+\frac{1}{4}$

4 If $\sin\theta + \cos\theta = \sqrt{3}/2$, and θ is an acute angle, find the value of $\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$. $(s-c)^2$ $1-2sc$

यदि $\sin\theta + \cos\theta = \sqrt{3}/2$ है, और θ एक न्यूनकोण है, तो $\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$ का मान ज्ञात कीजिए।

a) 1
b) 3/2
c) 5/4
d) 7/8

$(S+C)^3 - 3SC(S+C)$

5 If $sinA + cosA = \sqrt{2}$, then find $sin^3A + cos^3A$.

यदि $sinA + cosA = \sqrt{2}$, तो $sin^3A + cos^3A$ ज्ञात कीजिए।

a) $\frac{\sqrt{2}}{2}$

b) $\frac{\sqrt{2}}{5}$

c) $\frac{\sqrt{3}}{2}$

d) $\frac{\sqrt{5}}{2}$

$A = 45°$

$\sin + \cos = \sqrt{2}$

$\text{Cosec} + \sec = 2\sqrt{2}$

$\tan + \cot = 2$

$1 - 2s^2c^2$

$1 - 2 \times \frac{1}{2}$

$1 + 2sc = \frac{3}{2}$

6 If $\sin\theta + \cos\theta = \sqrt{2}$, where $0° < \theta < 90°$, find the value of $\sin^4\theta + \cos^4\theta$.

यदि $\sin\theta + \cos\theta = \sqrt{2}$ है, जहाँ $0° < \theta < 90°$ है, तो $\sin^4\theta + \cos^4\theta$ का मान ज्ञात कीजिए।

a) 1/2
b) 1/4
c) 3/4
d) 1

$\frac{1}{SC} = 2$

7 If $tanA + cotA = 2$ and $A \in (0, \pi/2)$, then what is sinA + cosA?

यदि $tanA + cotA = 2$ और $A \in (0, \pi/2)$, तो sinA + cosA क्या है?

a) 1

b) $\sqrt{2}$

c) $\frac{1}{\sqrt{2}}$

d) Cannot be determined

$(S+C)^2 = \sqrt{1+2SC}$

PARMAR SSC

①

$C + S = \frac{5}{4}$

$1 + 2SC = \frac{25}{16} - 1 = \frac{9}{16} \times 2$

$C - S = \sqrt{\frac{25}{16} - \frac{9}{8}} = \frac{\sqrt{7}}{4}$

$\cot = \frac{C}{S} = \frac{S + \sqrt{7}}{S - \sqrt{7}}$

8 If $cosA + sinA = \frac{5}{4}$, find tanA.

यदि $cosA + sinA = \frac{5}{4}$, tanA खोजें।

a) $\frac{5-\sqrt{7}}{5+\sqrt{7}}$

b) $\frac{4-\sqrt{7}}{4+\sqrt{7}}$

c) $\frac{5-\sqrt{2}}{5+\sqrt{2}}$

d) $\frac{2-\sqrt{3}}{2+\sqrt{3}}$

$(a-b)^2 = (a+b)^2 - 4ab$

$a + b = 9$

$a - b = 3$

$\frac{9+3}{2}$ $\quad$ $\frac{9-3}{2}$

① $(\sec^2+\csc^2)^2-2\sec^2\csc^2$

$52^2-2\times52$

$=52\times50$

$(\tan+\cot)^2=\tan^2+\cot^2+2$

$\sec^2+\csc^2=52$

$\frac{1}{c^2}+\frac{1}{s^2}$

$=\frac{s^2+c^2}{s^2c^2}=52$

$s^2+c^2=\sqrt{52}$

$\frac{1}{c^4}+\frac{1}{s^4}=\frac{s^4+c^4}{s^4c^4}$

$=\frac{1-2s^2c^2}{s^4c^4}$

9 If $\tan^2\theta + \cot^2\theta = 50$, then find $\sec^4\theta + cosec^4\theta$.

यदि $\tan^2\theta + \cot^2\theta = 50$ हो, तो $\sec^4\theta + cosec^4\theta$ का मान ज्ञात कीजिए।

a) 2600
b) 2800
c) 3200
d) 3000

a^4+b^4

$=(a^2+b^2)^2-2a^2b^2$

PARMAR
SSC